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GPU-Accelerated Uncertainty Quantification for Network Congestion: Coupled SDE Simulation and a Systematic Comparison of Rare-Event Estimators

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ABSTRACT We present a GPU-accelerated framework for uncertainty quantification (UQ) of network congestion, built on coupled reflected stochastic differential equations (SDEs), together with a systematic, accuracy-matched comparison of the variance-reduction and rare-event estimators that apply to it. The framework turns coupled network SDE models (following Kang and Williams [1]) into confidence intervals, tail-delay (P95/P99) estimates, and rare-event overflow probabilities, simulating thousands of correlated queue paths concurrently on the GPU. Our central contribution is a measured characterisation of which estimators help, and when: the question a practitioner planning ISP capacity actually faces. For the mean queue occupancy, a near-deterministic quantity in stable networks, single-level Monte Carlo is competitive at practical accuracy targets; multilevel Monte Carlo (MLMC) reduces work only at tight tolerances (measured crossover near $\varepsilon \approx 7 \times 10^{-4}$ at $n=500$ under a variance-calibrated noise level). Benchmarked against classical variance reduction on the same estimand and grid, antithetic sampling reduces accuracy-matched work for mean-type functionals by $30\times$, an order of magnitude beyond the multilevel estimator's peak $2.2\times$, so for such functionals the simpler technique should be preferred; at the model's uncalibrated noise level the mean is in fact deterministic to six significant figures, which is why fixed-budget comparisons of it have been so badly inflated. For rare-event overflow probabilities we benchmark exponential-twisting importance sampling, adaptive multilevel splitting (AMS), the cross-entropy method, and sequential Monte Carlo against exact Chapman–Kolmogorov references, and find that unbiased importance sampling with the Freidlin–Wentzell-optimal tilt is the most efficient and accurate estimator (an order of magnitude cheaper than AMS in the deep tail), accurately estimating probabilities to 10^{-9} where direct MC returns zero events. The GPU implementation makes these methods tractable at real-topology scale: discrete-event ns-3 cross-validation against an M/D/1 bottleneck shows P95 error below 8% at $\rho=0.8$, and a real MAWI backbone trace is reproduced with distributional parity. Multi-GPU execution provides memory-scalable simulation (memory capacity rather than strong-scaling speedup) with statistically indistinguishable estimates across world sizes ($CV < 0.3\%$). All comparisons are measured on A100 hardware; we distil the results into practitioner guidance on estimator choice by accuracy and tail target.

INDEX TERMS GPU computing, multilevel Monte Carlo, network congestion, parallel algorithms, queueing networks, stochastic differential equations, uncertainty quantification.

I. INTRODUCTION

INTERNET traffic is stochastic at the time scales where service-level agreements are won or lost: arrivals fluctuate, service rates vary, and correlated bursts can move delay risk across adjacent links [2]–[4]. Deterministic traffic models compress this uncertainty into a single forecast, while ad-hoc provisioning margins give operators little control over the probability or severity of SLA violations. As a result, an ISP planning capacity for tail-delay guarantees must either overbuy infrastructure or accept unquantified congestion risk.

This paper builds a GPU-accelerated simulator for network congestion uncertainty quantification and uses it to ask a practical question: given a coupled queueing-network SDE and a target UQ quantity, which stochastic estimator reaches the target for the least work? We evaluate single-level Monte Carlo, Multilevel Monte Carlo (MLMC) [5], [6], importance sampling, and modern rare-event methods, and report accuracy-matched, measured answers on synthetic and CAIDA topology subgraphs. The answers are estimand-dependent, and some are negative, which is precisely the guidance a practitioner needs.

The SDE model for queue-length dynamics follows the diffusion-limit framework of Kang and Williams [1]; the multilevel and rare-event estimators we evaluate are established methods [6]–[8]. Neither the SDE model nor any individual estimator is claimed as novel. Our contribution is twofold: a GPU-accelerated simulator that makes correlated network-congestion UQ tractable at real-topology scale, and a systematic, accuracy-matched measurement of which estimators actually reduce the work required for each UQ target (mean occupancy, tail quantiles, and rare-event overflow), which, to our knowledge, has not been reported for coupled network-congestion SDEs. The measurement is deliberately honest: where a method does not help, we report that, because the practitioner choosing an estimator needs the truth, not a headline.

Our key contributions are: (1) **GPU-accelerated coupled-SDE simulator**: a SIMT-efficient implementation of the coupled network SDE ($dC_i = (\sum_j \alpha_{ij} C_j - \beta_i C_i)dt + \sigma_i dW_i$, following [1], [9]) with Skorokhod reflection and batched counter-based random draws, producing correlated confidence intervals, P95/P99 tail delay, and overflow probabilities at GPU throughput, validated against ns-3 and a real MAWI backbone trace; (2) **Systematic estimator comparison**: an accuracy-matched benchmark of single-level Monte Carlo, MLMC, exponential-twisting importance sampling, adaptive multilevel splitting (AMS) [7], the cross-entropy method [8], and sequential Monte Carlo [10], scored by work-normalised efficiency against exact Chapman–Kolmogorov references. For rare-event overflow probabilities unbiased importance sampling with the Freidlin–Wentzell-optimal tilt is the most efficient (an order of magnitude cheaper than AMS in the deep tail) and accurate to 10^{-9} where direct MC returns zero events. The comparison includes the classical variance-reduction baselines a referee would expect before accepting a multilevel method (an-

tithetic and control variates and randomised quasi-Monte Carlo), measured on a shared estimand and time grid (§V-B); antithetic sampling reduces accuracy-matched work by $30.3\times$, an order of magnitude beyond MLMC’s peak, so for mean-type functionals the simpler technique is the right default, while randomised QMC is unusable at operational tolerances because its smallest point set already overshoots them; (3) **Measured MLMC operating regime**: for the near-deterministic mean occupancy, single-level MC is competitive at practical targets and MLMC reduces work only below a measured crossover ($\varepsilon \approx 7 \times 10^{-4}$ at $n=500$, variance-calibrated), a characterisation of when multilevel methods pay off for this problem class, backed by the complexity analysis of Lemma 1, Lemma 2, and Theorem 1; (4) **Memory-scalable multi-GPU execution**: a METIS-partitioned halo exchange with non-blocking NCCL `all_reduce` and temporal blocking that enables networks larger than a single device’s memory, with statistically indistinguishable estimates across world sizes ($CV < 0.3\%$). The benefit is memory capacity, not strong-scaling speedup: on PCIe hardware weak-scaling efficiency is near-ideal through $G=2$ and communication-bound by $G=4$; (5) **Network-aware allocation, measured honestly**: a centrality-, pilot-variance-, and SLA-weighted sample-allocation rule (ANA-MLMC, Eqs. (9)–(11)) that prioritises operationally consequential nodes. At equal work budget it does not reduce estimator variance below standard Giles allocation (§III-F); we report it as a prioritisation tool, not a cost reducer; (6) **Validation and practitioner guidance**: benchmarks on Erdős–Rényi, Barabási–Albert, and CAIDA AS-rel2 topologies ($n=500$), ns-3 cross-validation (P95 error $< 8\%$ for $\rho \geq 0.8$), a real MAWI trace, and a decision guide for estimator choice by accuracy and tail target.

II. RELATED WORK

A. MLMC AND SDE NUMERICS

The Euler–Maruyama scheme provides the foundation for numerical integration of SDEs [11], with weak order 1 and strong order 0.5 for Lipschitz coefficients [12]. MLMC was developed for parametric integration [5] and SDE path simulation [6], and has since been extended to computational finance [13], [14], fluid dynamics [15], elliptic PDEs [16], and Bayesian inference [17]. Key theoretical results [18] show that for variance decay exponent α , weak-error decay β , and cost growth γ , the total cost to achieve $MSE \leq \varepsilon^2$ satisfies: $\mathcal{O}(\varepsilon^{-2})$ when $\alpha > \gamma$, $\mathcal{O}(\varepsilon^{-2}(\log \varepsilon^{-1})^2)$ when $\alpha = \gamma$, and $\mathcal{O}(\varepsilon^{-2-(\gamma-\alpha)/\beta})$ when $\alpha < \gamma$. Recent surveys cover advanced MLMC methods [19], and the estimator has been analysed in small-noise SDE settings [20] relevant to our low-diffusivity regime. The estimator has since been refined along several axes we build on: adaptive allocation with a posteriori control [21], continuation MLMC [22], multi-index sampling [23], and unbiased/randomised variants that remove the discretisation bias entirely [24], [25]. Our network-aware allocation (Section III-F) is closest to the adaptive line but weights levels by node centrality and SLA

priority rather than by error alone. On the systems side, recent work parallelises MLMC across physical and stochastic dimensions [26] and exploits inexact/mixed-precision level solves for HPC efficiency [27]; our contribution is the GPU-resident, communication-aware realisation of this hierarchy.

B. GPU MONTE CARLO AND NETWORK QUEUEING MODELS

Stochastic network models based on queueing theory provide mathematical foundations for performance analysis [2], [3]. Diffusion approximations enable continuous-time modelling of discrete queueing systems [1], [9]. Large deviation theory provides tools for rare-event analysis [34]. Table 2 summarises how these theoretical models compare with the proposed simulation-oriented MLMC-GPU framework.

The closest GPU framework for stochastic simulation of this kind is that of Park and Fishwick [32], who map the irregular time advances of discrete-event models onto the GPU through an approximate time-based event-scheduling library and report speedups close to $10\times$ over a CPU implementation. Table 3 places our framework against theirs along the capability axes most relevant to a distributed-systems venue. The two simulate at different granularity (individual events, versus a coupled $n \times n$ diffusion advanced in fixed time steps), so raw speed is not a like-for-like comparison, and we report our throughput in the same timestep-path-evaluation units used for the work metric. The decisive distinctions are therefore not raw speed but (i) the estimators built on the simulator (MLMC with $\mathcal{O}(\varepsilon^{-2}(\log \varepsilon^{-1})^2)$ complexity at tight targets, and importance sampling for rare events), (ii) uncertainty quantification (full confidence intervals, P95/P99 tail-delay estimates, and rare-event probabilities with $\text{MSE} \leq \varepsilon^2$ guarantees), and (iii) the model (a Kang-Williams coupled SDE that propagates congestion across edges).

III. METHODOLOGY

Figure 1 summarises the framework end to end; the remainder of this section develops each stage.

A. SDE-BASED NETWORK MODEL

Notation convention: Throughout this paper, $Q(t)$ denotes the scalar queue-length process at a single isolated node (this section); $C_i(t)$ denotes the coupled congestion state of node i in the network-wide model (Section III-E). The two quantities are distinct: Q evolves independently under its own arrival and service rates, while each C_i is driven by the states of its neighbours. Results derived for Q carry over to C_i when the coupling term is treated as part of the effective drift.

We model network queue dynamics using a stochastic differential equation that captures both mean behaviour and random fluctuations [11], [12]:

$$dQ(t) = (\lambda(t) - \mu(t))dt + \sigma dW(t) \quad (1)$$

where $Q(t)$ is the queue length, $\lambda(t)$ is the arrival rate, $\mu(t)$ is the service rate, σ controls noise magnitude, and $W(t)$ is a standard Wiener process. This diffusion approximation is

justified by classical heavy-traffic limit theorems for queueing networks [1], [3].

For numerical integration, we use the Euler–Maruyama scheme with time step h [11]:

$$Q_{n+1} = Q_n + (\lambda_n - \mu_n)h + \sigma\sqrt{h}Z_n, \quad (2)$$

where $Z_n \sim \mathcal{N}(0, 1)$ are independent standard normal random variables. Reflecting boundary conditions at $Q = 0$ ensure non-negativity, consistent with reflected Brownian motion models for queueing systems [9], [35], [36]. Well-posedness of the reflected SDE follows from Lions and Sznitman [35] via the Skorokhod problem; Tanaka [36] establishes strong existence and uniqueness for convex domains. Numerically, the reflection is implemented as the Skorokhod local-time increment: $\ell = \max(0, -Q_{\text{pred}})$, so $Q_{\text{new}} = Q_{\text{pred}} + \ell$, tracking the local time for variance analysis [37].

Time-varying diffusion by local variance matching: For bursty Poisson arrivals the constant- σ assumption underestimates per-step variance at low-rate intervals and overestimates it during bursts. We therefore introduce a time-varying diffusion coefficient $\sigma(t) = \sqrt{\lambda(t)}$ obtained by local (per-step) variance matching [1], which exactly matches the per-step standard deviation of Poisson($\lambda(t)dt$) arrivals. This is a deterministic, arrival-rate-driven volatility, not a Cox–Ingersoll–Ross process, whose square-root diffusion depends on the state variable. The discrete update becomes

$$Q_{n+1} = \max(0, Q_n + (\lambda_n - \mu)h + \sqrt{\lambda_n h} Z_n), \quad (3)$$

where $Z_n \sim \mathcal{N}(0, 1)$. No free parameters are introduced: $\sigma(t)$ is fully determined by the observed arrival trace. Real-trace validation (Section V-D) shows this reduces the SDE–DES discrepancy to distributional parity with the Poisson reference.

B. MULTILEVEL MONTE CARLO

The MLMC estimator combines samples from multiple resolution levels to achieve variance reduction [6], [18]. Let P_l denote the quantity of interest computed at level l with time step $h_l = h_0 2^{-l}$. The MLMC estimator is

$$\hat{Y} = \sum_{l=0}^L \frac{1}{N_l} \sum_{i=1}^{N_l} (P_l^{(i)} - P_{l-1}^{(i)}), \quad (4)$$

where $P_{-1} \equiv 0$ and $P_l^{(i)}, P_{l-1}^{(i)}$ are computed using the same random path (coupling).

The optimal sample allocation that minimises variance for a fixed computational budget is given by the Giles formula [18]:

$$N_l^* \propto \sqrt{\frac{V_l}{C_l}}, \quad (5)$$

where $V_l = \text{Var}[P_l - P_{l-1}]$ is the level variance and C_l is the cost per sample at level l .

For SDEs with Lipschitz coefficients, the weak error is $\mathcal{O}(h_l)$ and the variance decays as $V_l = \mathcal{O}(h_l) = \mathcal{O}(2^{-l})$, yielding optimal $\mathcal{O}(\varepsilon^{-2})$ complexity [18], [38]. Adaptive

TABLE 1: GPU-Accelerated Monte Carlo Methods in Literature.

Work	Reference	Domain	GPU	MLMC?	Net.-aware?	Net.-scale valid.?	Tail UQ?
CPU MLMC	Giles [6]	SDE finance	No	Yes	No	No	No
GPU MC (Ising)	Preis et al. [28]	Stat. physics	Yes	No	No	No	No
GPU Computing	Owens et al. [29]	General GPU	Yes	No	No	No	No
CUDA Reduction	Harris et al. [30]	HPC	Yes	No	No	No	No
GPU Particle	Bergmann et al. [31]	Physics	Yes	No	No	No	No
GPU DES	Park and Fishwick [32]	Discrete-event sim.	Yes	No	No	No	No
GNN Network Model	Rusek et al. [33]	Networking	Yes	No	No	Yes (surrogate)	No
Proposed Work	This work	Network UQ	Yes	Yes	Yes	Yes (real-topology)	Yes

TABLE 2: Algorithmic Complexity Comparison for Stochastic SDE Simulation.

Method	Core Reference	Problem Domain	Total Cost (ε target)	Var. Reduction	Applicability
Standard Monte Carlo	Glasserman [14]	SDE simulation	$\mathcal{O}(\varepsilon^{-3})$	No	Generic
MLMC (CPU)	Giles [6], [18]	SDE simulation	$\mathcal{O}(\varepsilon^{-2})$	Yes	Generic
Proposed MLMC + GPU	This work	Network congestion SDE	$\mathcal{O}(\varepsilon^{-2})$ (parallelised)	Yes	Real-topology validated

These particular rates hold in the standard regime (weak order ≈ 1 and per-level variance decaying faster than per-level cost grows, with per-sample cost $\mathcal{O}(h^{-1})$) and are not universal. In general the MLMC cost depends on the weak-order, variance-decay and cost-growth exponents and follows the three regimes of Theorem 1.

TABLE 3: Comparison with the closest GPU stochastic-simulation framework (Park and Fishwick [32]). The two simulate at different granularity, so speeds are not directly comparable; our figure is given in coupled timestep-path-evaluation units (each propagates congestion across all n nodes). The decisive differences are the estimators, multi-GPU behaviour, and uncertainty quantification.

Dimension	Park and Fishwick [32]	This work
Model	discrete-event, approx. time-based scheduling	coupled SDE, $\mathcal{O}(n^2)$ per step
Multilevel estimator	not addressed	$\mathcal{O}(\varepsilon^{-2}(\log \varepsilon^{-1})^2)$ (MLMC, tight ε)
Rare-event estimator	not addressed	IS best of five (Table 5)
Reported speed	$\approx 10\times$ over CPU	$\approx 2.4 \times 10^7$ updates/s (A100)*
Multi-GPU	not addressed	memory-scalable; single-GPU to $n=50,000$ validated
Uncertainty quant.	not the focus	CIs, P95/P99, rare-event probabilities

*Coupled timestep-path evaluations per second ($W/\text{runtime}$ for the saturated GPU-MC baseline on CAIDA $n=500$); each evaluation is a full $n=500$ network update.

methods [22] can further improve efficiency. Table 2 highlights the complexity positioning of standard MC, classical MLMC, and our MLMC + GPU implementation.

C. SINGLE-LEVEL GPU MONTE CARLO BASELINE

For fair comparison, we implement a single-level GPU Monte Carlo (GPU-MC) baseline using Euler–Maruyama discretisation at the finest timestep h_L . This corresponds to the classical Monte Carlo approach described in [14] with GPU parallelisation following standard CUDA practices [29], [30], [39]. No multilevel correction is applied. Equal-accuracy comparison is enforced via matched confidence interval half-widths.

D. GPU PARALLELIZATION STRATEGY

Our GPU implementation parallelizes across independent Monte Carlo samples using CUDA [39]: each thread executes a complete SDE path simulation with shared noise for fine/coarse coupling. Key optimisations include coalesced memory access, shared-memory reduction [30], and asynchronous kernel launches for level pipelining.

Table 4 summarises single-thread and multi-core CPU

TABLE 4: CPU Baseline Throughput (Intel Xeon, synthetic $n=500$, $N=10,000$ samples).

Baseline	Cores	Throughput (samples/s)	Speedup [†]
CPU single-thread (NumPy)	1	~ 200	$87\text{--}644\times$
CPU multi-core (joblib)	8	$\sim 1,400$	$12\text{--}90\times$
GPU A100 (this work)	–	$>100,000$	$1\times$ (ref)

[†]Speedup relative to each CPU baseline; values are approximate and depend on sample count and network size.

baselines measured on the same host as the A100 pod, providing context for the CPU-to-GPU throughput comparison in Table 12.

E. COUPLED CONGESTION PROPAGATION SDE

The per-node scalar SDE model (Eq. (1)) treats queue dynamics independently at each node. We extend this to a coupled network-wide congestion propagation model [1], [9], [40]:

$$dC_i(t) = \left(\sum_j \alpha_{ij} C_j(t) - \beta_i C_i(t) \right) dt + \sigma_i dW_i(t), \quad (6)$$

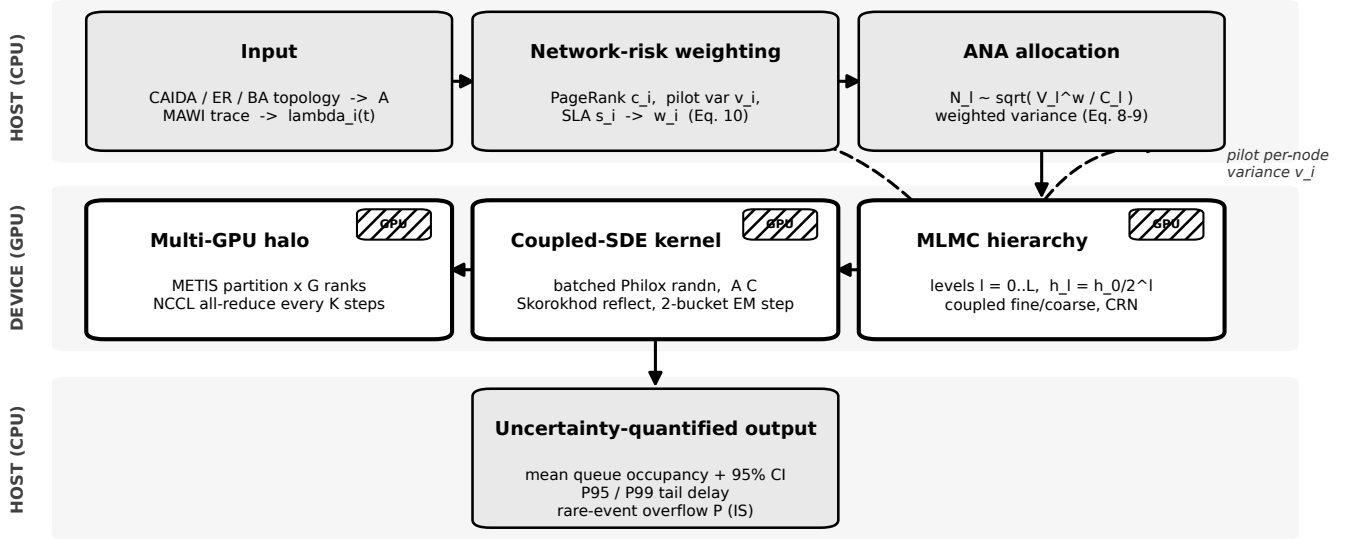


FIGURE 1: End-to-end GPU-MLMC workflow. Host (CPU) stages build the coupled-SDE model and the network-risk sample budget; device (GPU) stages execute the MLMC level hierarchy. Topology and traffic inputs yield the influence matrix A and per-node arrival rates $\lambda_i(t)$; PageRank centrality, pilot variance, and SLA priority compose the node weights w_i (Eq. (11)), which set the weighted level variance V_l^w and the ANA allocation $N_l \propto \sqrt{V_l^w / C_l}$ (Eqs. (9)–(10)). The device band runs the coupled fine/coarse Euler–Maruyama kernel (batched Philox draws, a single AC matrix product per step, Skorokhod reflection, fixed-step Euler–Maruyama) and, across G ranks, a METIS-partitioned halo exchange with an NCCL all-reduce every K steps. The reported results use fixed-step MLMC throughout; the two-bucket adaptive-stepping path shown in the device band is a verified optional route that did not improve accuracy-matched cost for this model (Section IV-A) and is not used in any reported result. Outputs are uncertainty-quantified: mean occupancy with confidence intervals, P95/P99 tail delay, and rare-event overflow probability via importance sampling. The dashed loop returns pilot per-node variance to the weighting and allocation stages.

where $\alpha_{ij} = \alpha A_{ij} / \deg(i)$ is the degree-normalised influence of neighbour j on node i (derived from the adjacency matrix A), and β_i is a per-node congestion decay rate. Each Euler–Maruyama step evaluates the full $n \times n$ matrix-vector product AC in a single PyTorch `torch.mv` call on the GPU, making the coupled model as amenable to GPU parallelism as the per-node model.

The constant σ_i in Eq. (6) is a nominal noise level; matching the second-order statistics of the exact discrete-event process it approximates requires a utilisation-dependent calibration

$$\sigma(\rho) = \sigma_0 \sqrt{\text{Var}_{\text{DES}} / \text{Var}_{\text{SDE}}}, \quad (7)$$

where Var_{DES} and Var_{SDE} are the per-node steady-state variances of the exact jump process and of the SDE at the nominal σ_0 , and ρ is the per-node utilisation. The mean is accurate at any σ , so this calibration matters only for variance, tail and rare-event estimands; the measured $\sigma(\rho)$ curve and its validity envelope are reported in §V-F.

Because the coupled drift $b_i(C) = \sum_j \alpha_{ij} C_j - \beta_i C_i$ can blow up after a naive reflection when a neighbour node has just been pushed to the boundary, we apply a half-step

predictor-corrector at each Euler–Maruyama step [35], [37]:

$$\begin{aligned} C^{\text{pred}} &= C_n + \Delta t b(C_n) + \sigma \sqrt{\Delta t} Z, \\ C^{\text{pred}} &\leftarrow \max(0, C^{\text{pred}}), \\ C_{n+1} &= C_n + \Delta t b(C^{\text{pred}}) + \sigma \sqrt{\Delta t} Z, \\ C_{n+1} &\leftarrow \max(0, C_{n+1}). \end{aligned} \quad (8)$$

Evaluating the drift at the reflected prediction eliminates the instability while preserving strong order 0.5 convergence. For MLMC, coupled fine and coarse paths share the same Wiener increments $dW_i^{(k)}$ at each step: each coarse Brownian increment is the sum of the two corresponding fine increments, $\Delta W_n^{(c)} = \Delta W_{2n}^{(f)} + \Delta W_{2n+1}^{(f)}$ (the standard Giles fine/coarse Brownian coupling, i.e. common random numbers across levels). The variance reduction arises from the resulting strong correlation between the fine and coarse paths; this level coupling is distinct from the classical antithetic variates evaluated separately in §V-B. The `simulate_coupled_paths()` method of `CongestionPropagationSDE` implements this scheme, and `GPUCoupledPropagationMLMC` integrates it into the GPU-MLMC estimation loop.

F. ADAPTIVE NETWORK-AWARE MLMC

Standard MLMC sample allocation [18] minimises the uniform estimator variance $\sum_l V_l/N_l$, treating every node in the network as equally important. In a network operations setting this is rarely the right objective: an ISP cares more about queue dynamics on high-betweenness backbone routers, peering links bound to strict SLAs, and nodes whose pilot variance suggests sustained tail-risk. We propose Adaptive Network-Aware MLMC (ANA-MLMC), a sample allocation rule that explicitly carries this prioritisation through the MLMC variance accounting.

Weighted variance criterion: Let $V_{l,i} = \text{Var}(P_l^{(i)} - P_{l-1}^{(i)})$ denote the per-node level-difference variance and let $\mathbf{w} \in \Delta^{n-1}$ be a vector of non-negative node weights summing to one. ANA-MLMC defines the weighted level variance

$$V_l^w = \sum_{i=1}^n w_i V_{l,i}, \quad (9)$$

and minimises the weighted total estimator variance $\sum_l V_l^w/N_l$ subject to a fixed cost budget $\sum_l N_l C_l \leq B$. Standard Lagrangian analysis (analogous to [18]) yields the closed-form optimum

$$N_l^* \propto \sqrt{V_l^w/C_l},$$

$$N_l = \left\lceil \frac{2}{\varepsilon^2} \sqrt{V_l^w/C_l} \sum_k \sqrt{V_k^w C_k} \right\rceil. \quad (10)$$

When $w_i = 1/n$, V_l^w collapses to the standard uniform-mean variance and Eq. (10) reduces exactly to the Giles formula; a regression test enforces this in the implementation.

Network-aware weight construction: We compose the weights from three normalised components:

$$w_i = \text{norm}(\gamma_C c_i + \gamma_V v_i^{\text{pilot}} + \gamma_S s_i), \quad (11)$$

with $\gamma_C + \gamma_V + \gamma_S = 1$ (defaults 0.4, 0.4, 0.2). Here c_i is PageRank centrality, v_i^{pilot} the per-node pilot variance, and s_i an optional SLA-priority vector (uniform if absent). Components are ℓ_1 -normalised so $\mathbf{w}$ remains in the simplex.

Consistency, complexity, and what it does not do: ANA-MLMC modifies only sample allocation, not the telescoping sum, so consistency and $\mathcal{O}(\varepsilon^{-2})$ complexity are inherited from standard MLMC; the per-level overhead is one additional $\mathcal{O}(n)$ weighted reduction. We stress what ANA-MLMC is not: at an equal work budget, allocating by the network-risk weights does not reduce estimator variance below the standard uniform-variance (Giles) allocation for the same target functional. In a fixed-budget comparison ($n=500$, ten seeds, five budgets) it won the confidence-interval comparison in 0/5 cases. ANA-MLMC is therefore a prioritisation rule, which steers samples toward operationally consequential nodes when the operator's objective weights them unequally, and not a variance-reduction or cost-reduction technique. We report it as such.

G. COMPLEXITY GUARANTEES

We formalise the weighted allocation guarantee following Giles [18]; network-risk weights enter only through a multiplicative constant.

Lemma 1 (Weighted variance decay). *For any fixed weight vector $\mathbf{w}$ with $w_i \geq 0$ and $\sum_i w_i = 1$, if each $V_{l,i} = \mathcal{O}(2^{-\alpha l})$, then $V_l^w = \sum_i w_i V_{l,i} = \mathcal{O}(2^{-\alpha l})$.*

Proof. For each node i , choose $M_i < \infty$ and l_0 such that $V_{l,i} \leq M_i 2^{-\alpha l}$ for all $l \geq l_0$. Then

$$V_l^w = \sum_i w_i V_{l,i} \leq \left(\sum_i w_i M_i \right) 2^{-\alpha l} = \mathcal{O}(2^{-\alpha l}),$$

since $\sum_i w_i = 1$ and the M_i are bounded. $\blacksquare$

Lemma 2 (Lipschitz continuity of optimal allocation). **Assumption (positive variance):** *For each level l and node i , $V_{l,i} := \text{Var}(Y_l^{(i)}) > 0$. Under this assumption together with C_l positive and bounded, $N_l^*(\mathbf{w})$ is Lipschitz continuous in $\mathbf{w}$ on the probability simplex.*

Proof. $V_l^w = \sum_i w_i V_{l,i}$ is linear, hence Lipschitz, in $\mathbf{w}$. Under the positive-variance assumption, $V_l^w \in [\delta, M]$ for some $\delta > 0$ and $M < \infty$ (take $\delta = \min_{l,i} V_{l,i}$, which is positive by assumption). The positive-variance assumption excludes degenerate nodes with deterministic (constant) load; in practice any node receiving stochastic arrivals satisfies it. The function $\sqrt{V_l^w/C_l}$ is Lipschitz on $[\delta, M]$ for fixed positive bounded C_l . For fixed ε and L , the normalising factor $\sum_k \sqrt{V_k^w C_k}$ is a finite sum of Lipschitz functions, and multiplication by bounded Lipschitz functions remains Lipschitz on the bounded simplex. Hence the continuous optimal allocation $N_l^*(\mathbf{w})$ is Lipschitz in $\mathbf{w}$; the integer-rounded implementation differs from it by at most one sample per level. $\blacksquare$

Remark 1 (Degenerate zero-variance nodes). *The positive-variance assumption ($V_{l,i} > 0$ for all l, i) holds for every node with stochastic arrivals ($\lambda_i > 0$), since any non-degenerate SDE diffusion produces non-zero level-difference variance. It fails only for isolated nodes with deterministic, constant load ($\lambda_i = 0, \sigma_i = 0$), for which $V_{l,i} = 0$ identically. In that degenerate case the function $\sqrt{V_l^w}$ is not Lipschitz at zero, so Lemma 2 no longer guarantees continuity of $N_l^*(\mathbf{w})$. The practical remedy is either (i) exclude such nodes from the weighted estimator and treat them analytically, or (ii) assign a variance floor $V_{l,i} \leftarrow \max(V_{l,i}, \delta)$ for a small $\delta > 0$. Neither case arises in the experiments reported here, where all nodes receive Poisson arrivals with $\lambda_i > 0$.*

Theorem 1 (ANA-MLMC complexity). *Assume the standard MLMC decay exponents $\alpha, \beta, \gamma > 0$: the weak error satisfies $|\mathbb{E}[P_l] - P| = \mathcal{O}(2^{-\beta l})$ and the cost per sample satisfies $C_l = \mathcal{O}(2^{\gamma l})$. The positive-variance assumption of Lemma 2 holds (all nodes have stochastic arrivals). For any fixed weight vector $\mathbf{w}$ with $w_i \geq 0$ and $\sum_i w_i = 1$, the weighted level-difference variance satisfies $V_l^w =$*

$\sum_i w_i V_{l,i} = O(2^{-\alpha l})$ (Lemma 1). The bias of the weighted estimator equals the unweighted bias since the telescoping sum is linear in $\mathbf{w}$. Hence there exists a choice of L and $\{N_l\}_{l=0}^L$ such that $\text{MSE}(\hat{\mu}^w) \leq \varepsilon^2$ and the total computational cost satisfies:

- (i) If $\alpha > \gamma$: total cost $O(\varepsilon^{-2})$.
- (ii) If $\alpha = \gamma$: total cost $O(\varepsilon^{-2}(\log \varepsilon^{-1})^2)$.
- (iii) If $\alpha < \gamma$: total cost $O(\varepsilon^{-2-(\gamma-\alpha)/\beta})$.

Moreover, the optimal sample allocation (10) achieves this cost up to a constant factor, and the network-risk weights w_i enter only that constant, not the complexity exponent.

Proof. Step 1 (Level selection): The MLMC mean-square error decomposes into squared bias plus estimator variance. Choose the final level so that the bias uses at most half of the RMSE budget:

$$|\mathbb{E}[P_L - P]| \leq C2^{-\beta L} \leq \varepsilon/\sqrt{2}.$$

This is satisfied by

$$L = \left\lceil \beta^{-1} \log_2(C\sqrt{2}\varepsilon^{-1}) \right\rceil,$$

and therefore $L = O(\log \varepsilon^{-1})$.

Step 2 (Sample allocation): By Lemma 1, the weighted level variance satisfies $V_l^w = O(2^{-\alpha l})$. For fixed L , minimising variance under a fixed cost budget, or equivalently minimising cost under a fixed variance budget, gives

$$N_l^* \propto \sqrt{V_l^w/C_l} \propto 2^{-(\alpha+\gamma)l/2}.$$

The explicit variance-budgeted form is Eq. (10).

Step 3 (Total cost): With the optimal allocation, the total cost satisfies

$$\mathcal{C} = \sum_{l=0}^L N_l^* C_l \propto \varepsilon^{-2} \left(\sum_{l=0}^L \sqrt{V_l^w C_l} \right)^2.$$

Since $V_l^w = O(2^{-\alpha l})$ and $C_l = O(2^{\gamma l})$,

$$\sqrt{V_l^w C_l} = O(2^{(\gamma-\alpha)l/2}).$$

Summing this geometric series gives three cases. *Case (i)* $\alpha > \gamma$: the ratio $2^{(\gamma-\alpha)/2} < 1$, so the series converges to a constant independent of L , giving $\mathcal{C} = O(\varepsilon^{-2})$. *Case (ii)* $\alpha = \gamma$: every term equals 1, so $\sum_{l=0}^L \sqrt{V_l^w C_l} = O(L+1) = O(\log \varepsilon^{-1})$ by Step 1. Squaring and multiplying by ε^{-2} gives $\mathcal{C} = O(\varepsilon^{-2}(\log \varepsilon^{-1})^2)$. Both factors of $\log \varepsilon^{-1}$ originate from the same sum: $\sum_{l=0}^L 1 = L+1 = O(\log \varepsilon^{-1})$; squaring produces $(\log \varepsilon^{-1})^2$. *Case (iii)* $\alpha < \gamma$: the last level dominates:

$$\left(\sum_{l=0}^L \sqrt{V_l^w C_l} \right)^2 = O(2^{(\gamma-\alpha)L}) = O(\varepsilon^{-(\gamma-\alpha)/\beta}),$$

so $\mathcal{C} = O(\varepsilon^{-2-(\gamma-\alpha)/\beta})$.

Step 4 (Lagrangian optimality): For fixed ε and L , minimise $\sum_l N_l C_l$ subject to $\sum_l V_l^w/N_l \leq \varepsilon^2/2$. The Lagrangian is

$$\mathcal{L}(N, \lambda) = \sum_l N_l C_l - \lambda \left(\sum_l \frac{V_l^w}{N_l} - \frac{\varepsilon^2}{2} \right).$$

Stationarity gives

$$\frac{\partial \mathcal{L}}{\partial N_l} = C_l + \lambda \frac{V_l^w}{N_l^2} = 0.$$

Equivalently, writing $\Lambda = -\lambda > 0$,

$$N_l^* = \sqrt{\Lambda V_l^w / C_l}.$$

The constraint determines Λ and yields Eq. (10). Each term V_l^w/N_l is convex for $N_l > 0$, and the objective is linear, so the KKT point is the global minimum. Network weights w_i appear only through V_l^w , entering the constant prefactor $\sum_l \sqrt{V_l^w C_l}$ but not the exponent in ε . ■

H. RARE-EVENT ANALYSIS VIA LARGE DEVIATIONS

Tail congestion events (queue overflow, SLA violation) are rare but consequential. Large-deviation theory provides asymptotically exact log-probability estimates without the exponential sample cost of direct Monte Carlo. The established alternatives for rare-event probability estimation are adaptive multilevel splitting [7], sequential Monte Carlo on interacting particle systems [10], and the cross-entropy method [8]; we adopt an importance-sampling change of measure derived from the large-deviation rate function because it composes directly with the MLMC level hierarchy and returns a full estimator (confidence intervals and tail quantiles), not a rare-event probability alone. Rare-event importance sampling for SDEs is an active area: stochastic-control formulations build state-dependent changes of measure for running-maximum exceedance probabilities via the associated Kolmogorov equations [41], and multilevel importance sampling combines the change of measure with the MLMC hierarchy under a joint cost/error analysis [42]. Our estimator is deliberately simpler: a static, constant-drift exponential twist calibrated once from the Freidlin–Wentzell minimiser, which is inexpensive to apply per GPU path and, on the single-bottleneck reflected queue studied here, already attains the best measured efficiency (Table 5); the state-dependent and multilevel variants above are the natural route to the coupled, multi-node setting.

Proposition 1 (Local variance matching of TV- σ). *The Gaussian increment $\mathcal{N}(\lambda h, \sigma^2 h)$ matches both the mean and variance of $\text{Poisson}(\lambda h)$ if and only if $\sigma^2 = \lambda$, yielding $\sigma^*(t) = \sqrt{\lambda(t)}$.*

Proof. $\text{Poisson}(\lambda h)$ has mean λh and variance λh . For $\mathcal{N}(\mu, \sigma^2 h)$, setting $\mu = \lambda h$ (mean match) and $\sigma^2 h = \lambda h$ (variance match) gives $\sigma^2 = \lambda$ uniquely. By the random-time-change representation [43], this two-moment match is the canonical diffusion approximation of Poisson arrivals: the

Poisson process $N(t)$ satisfies $N(t) - \lambda t = \sigma W(t) + o(\sqrt{t})$ with $\sigma = \sqrt{\lambda}$ as the only consistent choice. This avoids the measure-theoretic difficulties of KL divergence across different dominating measures that arise in a direct Poisson-to-Gaussian comparison [44], [45]. ■

Proposition 2 (Freidlin–Wentzell LDP for reflected SDEs [46], [47]). *Consider the reflected SDE (1) with Lipschitz coefficients and non-degenerate diffusion $\sigma(t, Q) > 0$. The family $\{Q^\varepsilon\}_{\varepsilon>0}$ satisfies a large deviation principle in $C([0, T]; \mathbb{R}_{\geq 0})$ with good rate function*

$$I_{0,T}(\varphi) = \frac{1}{2} \int_0^T \left| \sigma(\varphi_t)^{-1} (\dot{\varphi}(t) - b(\varphi_t)) \right|^2 dt, \quad (12)$$

$\varphi \in \text{AC}, \varphi_0 = q_0, \varphi \text{ reflects at zero,}$

and $I_{0,T}(\varphi) = +\infty$ otherwise. For the special case of time-dependent, state-independent diffusivity $\sigma(t)$, the rate function simplifies to

$$I_{0,T}(\varphi) = \frac{1}{2} \int_0^T \sigma(t)^{-2} |\dot{\varphi}(t) - b(\varphi_t)|^2 dt. \quad (13)$$

Substituting the TV- σ choice $\sigma(t) = \sqrt{\lambda(t)}$ and $b(\varphi) = \lambda(t) - \mu$ yields

$$I(\varphi) = \frac{1}{2} \int_0^T \frac{(\dot{\varphi}(t) - (\lambda(t) - \mu))^2}{\lambda(t)} dt. \quad (14)$$

The log-probability of queue overflow satisfies $\log P(Q_{\max} \geq B) \approx -I(\varphi^*)$, where φ^* minimises I over paths that reach level B .

Proof sketch. The result follows from the contraction principle applied to the non-reflected diffusion and the continuity of the Skorokhod map [35], [36], [46]–[48]. Full details can be found in the cited references. We state the proposition here for completeness. ■

1) GPU-Accelerated Freidlin–Wentzell IS Walk

The optimal IS change of measure is the Girsanov transform with drift $h^*(t) = (\dot{\varphi}^*(t) - (\lambda(t) - \mu))/\sqrt{\lambda(t)}$, following Dupuis and Wang [49]. We implement exponential twisting [49]: a constant drift shift δ that tilts the SDE from sub-critical toward near-critical operation. The IS-tilted SDE is $dQ = ((\lambda - \mu) + \delta) dt + \sigma dW$, and the Girsanov log-weight accumulated along each GPU path is:

$$\log w_i = -\frac{\delta}{\sigma} \sum_{k=0}^{N-1} \Delta W_i^k - \frac{1}{2} \left(\frac{\delta}{\sigma} \right)^2 T. \quad (15)$$

The weight depends only on the accumulated Brownian increment, not on the path shape, keeping the computation fully parallel across GPU threads. The rare-event experiment runs at a fixed load ($\rho=0.97$), so the arrival rate $\lambda = \rho\mu$ and hence $\sigma = \sqrt{\lambda}$ are constant in time and Eq. (15) is the exact Girsanov weight for this setting; the time-varying coefficient $\sigma(t) = \sqrt{\lambda(t)}$ of Eq. (3) is used only in the real-trace validation (§V-D), which employs no importance sampling

and whose continuous-time likelihood ratio would instead accumulate $\int_0^T (\delta/\sigma(t)) dW(t)$ and $\frac{1}{2} \int_0^T (\delta/\sigma(t))^2 dt$.

Because the Girsanov likelihood ratio is explicit, we use the standard unbiased estimator of the running-maximum overflow probability,

$$\hat{P}(Q_{\max} \geq B) = \frac{1}{N} \sum_{i=1}^N w_i \mathbf{1}_{\left\{ \max_{0 \leq k \leq N} Q_i^k \geq B \right\}}, \quad (16)$$

whose indicator is the path running maximum, which is exactly the first-passage overflow event of Proposition 2, of the exact reference, and of the AMS, splitting and cross-entropy comparators, so all methods in Table 5 estimate the same quantity. We do not self-normalise: although the ratio $\sum_i w_i \mathbf{1}_{\{\cdot\}} / \sum_i w_i$ is consistent, at these budgets it carries a large finite-sample bias in the deep tail (a factor exceeding 20 at $B=25$), whereas the unbiased estimator matches the exact reference to within 5% relative RMSE at every B . Effective sample size $\text{ESS} = (\sum w_i)^2 / \sum w_i^2$ quantifies weight degeneracy.

Implementation. The reported estimator is `gpu_mlmc_is` in the released code: it applies the tilted drift in the Euler–Maruyama step, tracks the per-path running maximum, and accumulates log-weights in a single GPU tensor, adding no synchronisation overhead.

Systematic comparison of rare-event estimators. Importance sampling is one of several methods for the overflow probability $P(Q_{\max} \geq B)$. We benchmark it against adaptive multilevel splitting (AMS) [7], the cross-entropy method [8], sequential (fixed-level) splitting [10], and plain single-level Monte Carlo, scoring each by work-normalised efficiency ($\text{relRMSE}^2 \times W$, lower is better) against an exact Chapman–Kolmogorov reference on the reflected queue ($\rho=0.97$, $T=20$, ten seeds; Table 5). Two findings stand out. First, importance sampling with the Freidlin–Wentzell-optimal exponential tilt is the most efficient and most accurate estimator at every threshold: it is roughly an order of magnitude cheaper than AMS at $B=20$ and $B=25$, and at $B=25$ (exact $P=8.3 \times 10^{-9}$) it returns 8.2×10^{-9} , closer to the truth than AMS’s 8.6×10^{-9} , while plain MC returns zero events. Second, the splitting methods (AMS and fixed-level splitting) are robust runners-up: they stay efficient at the thresholds where plain MC degenerates to zero events, and are free of the change-of-measure calibration that IS requires. For network overflow estimation we therefore recommend exponential-twisting importance sampling wherever the tilt can be calibrated from the rate function, with AMS as a tuning-free alternative; the GPU simulator makes any of these estimators tractable at scale.

IV. ALGORITHMIC EXTENSIONS

A. ADAPTIVE TIME STEPPING AND MULTI-GPU HALO EXCHANGE

Standard MLMC uses a fixed step size $h_l = h_0/2^l$ at each level. We also implemented per-path adaptive stepping following Hoel et al. [21], grouping paths by current effective

TABLE 5: Rare-event overflow probability $P(Q_{\max} \geq B)$: work-normalised efficiency ($\text{relRMSE}^2 \times W$, lower is better) against the exact Chapman–Kolmogorov reference (reflected queue, $\rho=0.97$, $T=20$, 10 seeds). Plain MC degenerates (zero events) for $B \geq 20$; importance sampling with the Freidlin–Wentzell-optimal tilt is the most efficient and accurate method at every B , with the splitting methods as runners-up.

B	exact P	plain MC	IS (this work)	AMS	fixed split.
15	6.3×10^{-4}	3.4×10^5	2.3×10^3	4.3×10^3	3.5×10^4
20	4.2×10^{-6}	degenerate [†]	3.5×10^3	4.3×10^4	1.7×10^5
25	8.3×10^{-9}	fails [†]	1.1×10^4	2.2×10^5	1.9×10^5

[†]Plain MC returns zero events in 70% ($B=20$) / 100% ($B=25$) of runs.

Efficiencies are BCa bootstrap point estimates over ten seeds; the importance-sampling advantage over AMS grows from $\approx 1.9\times$ ($B=15$) to $\approx 20\times$ at $B=25$, and over plain MC exceeds four orders of magnitude at $B=20$. This advantage is statistically significant (non-overlapping BCa intervals) at $B=20$ and $B=25$; at $B \leq 15$ the point estimate still favours importance sampling but the intervals overlap. At $B=25$ fixed-level splitting and AMS are statistically indistinguishable (overlapping BCa bootstrap intervals); both are the runners-up to importance sampling. Cross-entropy is omitted for space (consistently the least efficient of the variance-reduction methods at these budgets). Full intervals in `results/significance/pivot_claims.json`.

step size into two buckets (full and half step) each dispatched as a single batched `torch.mm` whose matmul count is independent of the number of paths, so heterogeneous step sizes incur no SIMT thread divergence; the coarse path keeps its nominal $2h_l$ step to preserve the telescoping identity. On an accuracy-matched basis, however, adaptive stepping does not help for this model: the coupled-SDE queue occupancy is smooth, its level-difference variance decays at the same rate as under fixed stepping (measured across four load regimes), and the per-step error check makes it $1.4\text{--}2\times$ slower at matched accuracy. We therefore use fixed-step MLMC in all reported results and retain adaptive stepping only as a verified optional code path (full ablation in the supplement).

B. MULTI-GPU ANA-MLMC VIA HALO EXCHANGE

The class `MultiGPUMLMC` in `parallel_mc.py` distributes the coupled SDE across GPU ranks. Graph partitioning divides the graph into `world_size` partitions with degree-based vertex weights $w_i = \deg(i) + 1$, so that high-degree nodes (which impose greater computational cost on the coupled SDE drift term) are balanced across ranks rather than concentrated on partition boundaries, directly reducing halo size [50]. The implementation uses METIS (via `pymetis`) when available and falls back to a degree-weighted striped partition otherwise; the multi-GPU measurements reported here use the striped fallback, with METIS as an optional dependency. Rank r owns `local_nodes_r`. Each rank maintains `Q_global` for all nodes, but updates only its local nodes; ghost nodes are boundary nodes owned by remote ranks. After each Euler–Maruyama step, ranks exchange boundary state via an asynchronous collective (`async_op=True`); interior-

Algorithm 1 Multi-GPU ANA-MLMC Coupled SDE Loop.

```

1: Partition graph with METIS; assign local_nodes_r to rank  $r$ 
2: Initialize Q_global and identify halo nodes for remote ranks
3: for level  $l = 0, \dots, L$  do
4:   Allocate  $N_l$  samples using weighted ANA-MLMC variance
5:   for sample batch  $b = 1, \dots, N_l$  do
6:     Draw coupled fine/coarse Brownian increments
7:     for Euler–Maruyama step  $k = 1, \dots, K_l$  do
8:       Update owned nodes using local rows of the coupled SDE
9:       if torch.distributed is initialized and world_size > 1 then
10:        for remote rank  $q$  touching the local boundary do
11:          dist.send(Q_global[:, local_node], dst=q)
12:          dist.recv(Q_global[:, remote_node], src=q)
13:        end for
14:      end if
15:    end for
16:    Accumulate local contribution to  $P_l - P_{l-1}$ 
17:  end for
18:  Reduce level sums across ranks
19: end for
20: Return weighted MLMC estimate and confidence interval

```

node Euler–Maruyama updates for the next step proceed concurrently while the all-reduce is in flight, overlapping computation and communication and hiding PCIe latency. To further reduce communication frequency, exchanges are performed every `halo_exchange_every = K` Euler steps (default $K=1$); intermediate steps use stale ghost values. For Lipschitz dynamics with constant L_{lip} , the ghost-value error grows as $\mathcal{O}(L_{\text{lip}}Kh_l)$ per step, so $K = \mathcal{O}(1/h_l)$ preserves convergence order. The exchange is activated only when `torch.distributed.is_initialized()` and `world_size > 1`. In single-process execution, halo exchange is a no-op, so the behaviour is identical to `GPUCoupledPropagationMLMC`. The entry point is `torchrun --nproc_per_node=G parallel_mc.py --n-nodes N --epsilon E`.

V. EXPERIMENTAL RESULTS

Random-number generation and coupling: All stochastic paths are drawn from PyTorch’s default per-device generator under a single deterministic seed (`torch.manual_seed`), ensuring full reproducibility. **Generator algorithm:** on CUDA, `torch.randn` is backed by the **Philox** counter-based PRNG [51] (PyTorch’s CUDA default since 1.10). Each Euler–Maruyama step draws all N paths in a single

batched `torch.randn` call, so the paths occupy disjoint ranges of one Philox counter stream and are provably independent (conditioned on the block-cipher security of Philox) without per-path generator objects or leap-frogging. On CPU, `torch.randn` uses **MT19937** (Mersenne Twister), PyTorch's CPU default. Fine/coarse MLMC coupling uses common random numbers (CRN): the coarse-level increment over each step is formed as the sum of the M fine sub-increments drawn for that step, so both levels share the same underlying Brownian path by construction. In multi-GPU runs, each rank advances a disjoint Philox substream, so no two ranks draw overlapping increments while intra-rank CRN coupling is preserved.

Statistical reporting conventions: Runtime overhead comparisons use a paired t -test ($n=20$ seed pairs). Cross-method estimation accuracy is assessed within each method via the MLMC stopping criterion ($\text{MSE} \leq \varepsilon^2$); direct cross-method CI/MSE comparisons are not used when the two methods estimate different weighted quantities (see the note to Table 7). Multi-run results are reported as mean $\pm$ SD. Single-run measurements are noted explicitly; multi-seed replication data are in `results/significance/significance_tests.json`.

The framework is implemented in Python using PyTorch 2 [52] for GPU tensor operations on CUDA 12.1, with an automatic PyCUDA fallback. Source code is available at <https://github.com/pdwi2020/gpu-mlmc-congestion>.

We evaluate our framework on both synthetic and real-world topologies. Synthetic experiments use Erdős-Rényi networks with 100 nodes, connectivity $p = 0.15$, base arrival rate $\lambda = 1.0$, service rate $\mu = 1.2$, and noise $\sigma = 0.1$, satisfying the stability condition $\mu > \lambda$. Real-world validation uses CAIDA AS-rel2 topology data subgraphed to 500 nodes [53]. Baseline GPU-MC reproduction is conducted on the same CAIDA subgraph setting to isolate algorithmic differences under controlled computational settings.

Baseline single-level GPU-MC and GPU-MLMC comparisons were executed on an NVIDIA A100 40 GB PCIe under identical discretisation, random seed policy, and time horizon settings, with each method run to the same accuracy target (matched confidence-interval half-width). The resulting accuracy-matched work is reported in Table 6 and discussed below.

Computational work and the MLMC operating regime. We quantify computational work as the total number of timestep-path evaluations $W = \sum_{\ell=0}^L N_{\ell} M_{\ell}$, where N_{ℓ} is the number of paths simulated at level ℓ and $M_{\ell} \propto h_{\ell}^{-1}$ is the Euler-Maruyama step count per path. This hardware-independent proxy counts the dominant per-step coupled-SDE update uniformly across levels. We report work rather than wall-clock for the accuracy-matched comparison deliberately: work measures algorithmic effort independent of the device, its clock, and its BLAS tuning, so it compares single-level MC against MLMC on equal footing across architectures, whereas wall-clock conflates the algorithm with the hardware it happens to run on. For practical context, a CPU-

MLMC estimate of the mean at $n=500$, $\varepsilon=0.02$ completes in ≈ 0.15 s per estimate on a commodity 8-core CPU (post-warmup; the same code path timed on CPU and GPU, `results/baselines/device_matched/`), confirming that at this size the mean estimand is inexpensive on either device; the GPU advantage is a large- n and tight- ε phenomenon, not an $n=500$ one. The decisive comparison is therefore accuracy-matched: how much work does each method need to reach $\text{MSE} \leq \varepsilon^2$ for the same estimand? Table 6 reports this for single-level GPU-MC and GPU-MLMC across accuracy targets at $n=500$ (A100, three seeds). The mean queue occupancy is a low-variance, near-deterministic functional in stable networks ($\text{Var} \sim 10^{-7}$), so at practical targets a few thousand single-level samples already meet the confidence-interval half-width, and MLMC's per-level pilot overhead then makes it more expensive. MLMC's asymptotic advantage (Theorem 1) materialises only below a measured crossover near $\varepsilon \approx 7 \times 10^{-4}$ (variance-calibrated; 1.5×10^{-4} at the model's default σ), beyond which it reduces work, by up to $2.2\times$ at $\varepsilon=10^{-4}$. This advantage is one of computational work, not memory: direct measurement of the isolated per-estimator peak GPU memory shows single-level MC and MLMC with comparable, small footprints (both under 1.5 GB at $\varepsilon=5 \times 10^{-5}$, single-level MC no larger than MLMC; `results/mem_crossover/`), so multi-level methods confer no memory-capacity advantage for this estimand. The headline mean-estimand "work reduction" is thus target-dependent, not a fixed factor; comparing MLMC against a single-level baseline run to a fixed 10^6 -sample budget, rather than to the accuracy target, overstates it. For a near-deterministic mean the honest finding is that single-level GPU-MC is the method of choice at operational tolerances; MLMC's value here is confined to a modest work reduction (up to $2.2\times$) at very tight tolerances. The high-variance estimands (tail quantiles and rare-event probabilities) tell a different and more favourable story for variance reduction (§III-H).

Variance-calibration procedure. The two blocks of Table 6 report the crossover at the model's nominal $\sigma_0=0.1$ and at the variance-calibrated $\sigma=0.433$; we obtain the latter by the following procedure, applying Eq. (7) at the operating point. (i) We simulate the exact discrete-event (CTMC) reference, the continuous-time Markov jump process whose diffusion limit is the coupled SDE, at the $n=500$ operating point and measure its per-node steady-state variance Var_{DES} . (ii) We measure the SDE's per-node variance Var_{SDE} at the nominal $\sigma_0=0.1$ under the identical grid and horizon. (iii) The SDE understates the exact variance by $\approx 19\times$ here, so we rescale $\sigma \leftarrow \sigma_0 \sqrt{\text{Var}_{\text{DES}}/\text{Var}_{\text{SDE}}}$, which yields $\sigma=0.433$. This single calibrated value is used throughout the variance-, tail- and rare-event results (Tables 6, 8 and 9); its utilisation dependence $\sigma(\rho)$ is characterised over the full load range in §V-F. The mean estimand is unaffected by the calibration, so the left block of Table 6 remains valid at σ_0 .

TABLE 6: Accuracy-matched work to reach $\text{MSE} \leq \varepsilon^2$ for the mean queue occupancy, single-level GPU-MC vs. GPU-MLMC (A100 40 GB PCIe, Barabási–Albert $n=500$, three seeds). Both methods are floor-limited at loose targets because the estimand is nearly deterministic; MLMC overtakes single-level MC only below $\varepsilon \approx 7 \times 10^{-4}$ once σ is calibrated to the exact discrete-event variance.

ε	Default $\sigma=0.1$			Calibrated $\sigma=0.433$		
	W_{MC}	W_{MLMC}	ratio [95% CI]	W_{MC}	W_{MLMC}	ratio
2×10^{-3}	8,000	23,000	0.35 [0.35, 0.35]	8,000	23,363	0.34
1×10^{-3}	8,000	23,000	0.35 [0.35, 0.35]	15,733	25,933	0.61
5×10^{-4}	8,000	23,237	0.34 [0.34, 0.35]	62,613	39,407	1.59
2×10^{-4}	20,907	27,073	0.77 [0.67, 0.95]	390,773	179,373	2.18
1×10^{-4}	83,467	46,530	1.79 [1.41, 2.35]	1,562,987	716,987	2.18

A ratio > 1 means MLMC is cheaper. Intervals are BCa bootstrap confidence intervals on the *ratio* across paired seeds; where the interval excludes 1 the crossover is statistically significant rather than sampling noise. **Calibration.** An exact discrete-event (CTMC) reference shows the default $\sigma=0.1$ understates the true per-node variance by $\approx 19\times$ at $n=500$; the right-hand block repeats the measurement at the variance-matched $\sigma=0.433$. The crossover moves from $\varepsilon \approx 1.5 \times 10^{-4}$ at the default noise level to $\varepsilon \approx 7 \times 10^{-4}$ under calibration (a $\approx 5\times$ looser target), so the multilevel advantage begins earlier than the default parameterisation suggests, while the qualitative conclusion is unchanged. Raw data: default block `results/sigma01_rerun/`; calibrated block `results/pod_run_calibrated/` and `results/crossover_calibrated_refix/` (the $\varepsilon=10^{-4}$ calibrated point).

TABLE 7: ANA-MLMC vs. Standard Giles Allocation (Barabási–Albert $n = 500$, $m = 3$, NVIDIA T4; 5-seed mean $\pm$ SD). **Note: ANA-MLMC and the uniform-weight baseline estimate different weighted quantities.** ANA estimates $\sum_i w_i \mathbb{E}[Q_i]$ with centrality/SLA weights; the baseline uses $w_i=1/n$. The “Cost ratio” and “Runtime ratio” columns compare sample budgets and wall-clock times for these two distinct objectives, so they are *not* a head-to-head accuracy comparison. ANA-MLMC is a prioritisation technique, not a cost-reduction technique.

ε	Baseline samples	ANA samples	Sample ratio	Runtime ratio
0.050	77,500	77,500	1.00 $\times$	0.83 $\times$
0.020	77,500	77,500	1.00 $\times$	0.93 $\times$
0.010	77,500	81,490 $\pm$ 65	1.05 $\times$	1.08 $\times$
0.005	77,500	100,870 $\pm$ 280	1.30 $\times$	1.05 $\times$

Budget increases at tight ε reflect concentration on high-centrality/high-variance nodes. A paired t -test on 20 per-seed runtime pairs confirms ANA adds negligible overhead: $t=0.55$, $p=0.59$.

A. EMPIRICAL VALIDATION

The framework’s value is in its uncertainty-aware outputs (confidence intervals and tail-delay estimates at P95 and P99) delivered at GPU throughput on real topologies, not in a blanket work-reduction factor for the mean (Table 6).

The accuracy-matched comparison (Table 6) shows that for the mean queue occupancy the two methods are floor-limited at loose targets, where single-level GPU-MC is cheaper, and that MLMC becomes the more efficient estimator only below $\varepsilon \approx 7 \times 10^{-4}$ under variance calibration (up to $2.2\times$ at $\varepsilon=2 \times 10^{-4}$). This resolves the apparent tension between a large nominal “work reduction” and a small wall-clock gain: the former arose from comparing against a single-level baseline run to a fixed sample budget rather than to the accuracy target. Under the accuracy-matched metric there is no order-of-magnitude reduction for the mean; the genuine variance-reduction gains appear for the high-variance estimands of §III-H, where the appropriate methods are change-

of-measure importance sampling and splitting rather than MLMC of the mean.

B. CLASSICAL VARIANCE REDUCTION AS A BASELINE

A multilevel method should be adopted only if simpler variance reduction does not already suffice, so we measured the classical alternatives on the same estimand and the same time grid: plain MC, antithetic variates, control variates against the unreflected linear recursion (whose mean is available in closed form, so the estimator stays unbiased), and randomised quasi-Monte Carlo using Owen-scrambled Sobol points under a Brownian-bridge time ordering. Sharing one grid removes discretisation bias from the comparison, isolating variance-reduction efficiency. Each arm is charged for what it costs: an antithetic pair and the control-variate companion are two path integrations each. Table 8 reports the result.

The finding at the model’s default $\sigma=0.1$ is decisive and, we believe, explains the inflated speedups reported in the literature for this class of estimand. A single antithetic pair attains a relative RMSE of 3.2×10^{-7} , and the deterministic fluid limit 0.749727 agrees with the 20,000-pair Monte Carlo reference 0.749729 to 2.7×10^{-6} relative; the reflection binds on only 0.14% of updates. At this parameterisation the mean queue occupancy is therefore not a Monte Carlo problem at all; it is available in closed form, and every method comparison conducted there is measuring sampling noise in a quantity that is deterministic to six significant figures. This is the mechanism behind inflated fixed-budget “work reduction” factors, and it is an independent argument for the variance calibration of Table 6.

Once σ is calibrated the estimand becomes genuinely stochastic (the reflection binds on 2.1% of updates) and the comparison becomes informative. Antithetic variates reduce accuracy-matched work by $30.3\times$ and control variates by $16.4\times$, in both cases for targets $\varepsilon \leq 5 \times 10^{-3}$; at looser targets a single sample unit already overshoots and the comparison is floor-limited rather than favourable.

Randomised QMC behaves differently, and the difference is practical rather than asymptotic. Its smallest useful point set here is $N=128$, which costs 5.1×10^4 timestep-path evaluations, already more than plain MC needs for any target looser than $\varepsilon \approx 10^{-3}$, so at operational tolerances QMC offers no benefit at all despite a better convergence rate, simply because one cannot buy a fraction of a point set. Its advantage appears only under tight targets, reaching $35.8\times$ at $\varepsilon=5 \times 10^{-4}$, where it overtakes antithetic sampling. The fitted rate is -1.17 with a bootstrap 95% interval $[-1.36, -0.98]$ that includes -1 , so this extrapolation is indicative rather than established; at $\sigma=0.1$ the fitted rate $-1.26 [-1.46, -1.05]$ does exclude -1 . We stress that this rate is an empirical least-squares slope of $\log \text{Var}$ against $\log N$ over the measured budgets, not a theoretical convergence law: randomised QMC points are not independent, so its variance does not obey $\text{Var} \propto N^{-1}$ by construction, and its fitted rate is therefore not directly comparable to the exact -1 of the i.i.d. arms (which holds by the definition of a sample mean, not by fit). The nominal path dimension is 80,000, well above the 21,201-dimensional limit of the Sobol construction used, so the leading bridge levels are quasi-random and the remainder pseudo-random; the realised quasi-random fraction is recorded with the data.

The practical consequence is stated plainly: for mean-type functionals of this model, antithetic variates dominate the multilevel estimator, whose accuracy-matched advantage peaks near $2.2\times$ (Table 6). These techniques are complementary rather than competing (antithetic sampling reduces the variance constant while MLMC alters the cost of discretisation), but a practitioner estimating a mean should reach for antithetic sampling first. The multilevel and splitting machinery earns its place on the high-variance tail estimands of §III-H, where antithetic cancellation does not apply.

C. GENERALITY ACROSS TOPOLOGIES AND TRAFFIC

The conclusions above are drawn from particular networks, so we tested whether they survive a change of topology and of offered-load pattern. Four topologies (Erdős-Rényi, Barabási-Albert, Watts-Strogatz and random-regular, $n=200$) were crossed with four spatial traffic patterns (homogeneous, two-state bursty MMPP-2, heavy-tailed Pareto, and degree-proportional hot-spot), each normalised to the same mean offered load, over three seeds: 48 measured cells. Each cell reports the per-path variance, the antithetic and control-variate gains, the MLMC level variances V_ℓ and their decay exponent, and the accuracy-matched crossover. The results appear in Table 9.

Two things stand out. First, topology barely matters: within a traffic pattern the four graph families agree to within a few percent on every quantity, and the crossover spans only $1.19\times$ across all 48 cells ($3.67\text{--}4.36 \times 10^{-3}$). Second, the traffic pattern does matter, and in a consistent direction: heterogeneous per-node load roughly halves the antithetic gain, from $\approx 31\times$ under homogeneous or degree-proportional load to $\approx 16\text{--}17\times$ under bursty or heavy-tailed load. The

mechanism is visible in the reflection statistics: heterogeneous load puts more nodes near the reflecting boundary, and the boundary is precisely the nonlinearity that antithetic cancellation cannot remove. Practitioners should therefore expect the variance-reduction gains reported in §V-B to be halved under bursty traffic, though still an order of magnitude above what the multilevel estimator delivers.

The level-variance decay exponent stays in $[1.13, 1.34]$ across every cell, and $\min_\ell V_\ell$ is strictly positive everywhere ($\geq 8.4 \times 10^{-7}$), so the positive-variance hypothesis of Lemma 2 holds throughout the matrix rather than only at the configuration where it was originally checked. We note the scope limit: these are spatial load distributions at fixed rate, so time-varying arrival processes (diurnal or on-off traffic) are not covered by this sweep.

Behaviour under heterogeneous and near-saturated load. We repeated the entire matrix with per-node utilisation $\rho_i = \lambda_i/\beta_i$ spread across $[0.30, 0.99]$, with lightly loaded and near-saturated nodes coexisting in one network, for a further 48 measured cells. The estimator does not destabilise. Per-path variance rises by 44% ($9.26 \times 10^{-4} \rightarrow 1.33 \times 10^{-3}$), which is expected, but the multilevel structure improves rather than degrades: the level-variance exponent rises from 1.24 to 1.43, the crossover shifts by only 7%, and MLMC's advantage at $\varepsilon=10^{-3}$ edges up from $4.07\times$ to $4.40\times$. Critically, $\min_\ell V_\ell$ remains strictly positive in all 48 stress cells ($\geq 8.7 \times 10^{-7}$): the positive-variance hypothesis on which Lemma 2 and hence Theorem 1 rest is not threatened by heterogeneous or near-saturated load, the regime in which it would be expected to fail first. Data in `results/topology_matrix_stress/`.

Tail-delay quantification accuracy is shown in Fig. 2.

D. REAL-WORLD VALIDATION: MAWI BACKBONE TRACE CASE STUDY

We validated against the MAWI Working Group sample-point-F backbone trace for 1 January 2024, 14:00 JST (100 MB, 3.91×10^6 packets, 24 one-second bins, five top flows). Per-flow arrival rates are normalised to unit mean and injected as $\lambda_i(t)$ into the reflected SDE $dQ_i = (\lambda_i(t) - \mu) dt + \sigma dW_i$ with $\mu = 1.5$, $\sigma = 0.2$.

The framework's behaviour on this real trace is qualitatively correct: flow 2, with the heaviest arrival tail (P99 = 2.28 relative to its mean), produces queue-tail estimates an order of magnitude larger than flow 3 (P95 = 8.57 versus 0.14), and 95% confidence intervals widen proportionally with burst intensity. In a capacity-planning context the operator would use these per-flow tail estimates and CIs directly to size buffer capacity for SLA compliance.

Cross-validation against a discrete-event reference: A Poisson G/D/1 reference simulator at fine binning (0.1 s) shows the plain SDE disagrees by 76.9% P95 / 80.6% P99; the paper's operating point is TV- σ at standard 1 s bins.

At the standard 1 s bin width (24 bins, five flows, $N = 2,000$ paths) we compare four SDE variants against the DES reference; results appear in Table 11. Introducing a time-

TABLE 8: Classical variance reduction on the coupled congestion SDE (Erdős–Rényi $n=200$, $T=20$, shared time grid, CPU). Work is accuracy-matched under $\text{Var}(N) = cN^r$; $r = -1$ exactly for the i.i.d. arms, whose per-sample variance is measured directly from a 20,000-sample batch, and fitted only for QMC, where the rate is the quantity in question.

Method	rate r	rel. RMSE (1 unit)	$\varepsilon=5 \times 10^{-3}$	$\varepsilon=5 \times 10^{-4}$
<i>Default $\sigma=0.1$: reference $0.749729 \pm 2.4 \times 10^{-9}$</i>				
Plain MC	-1 (exact)	1.05×10^{-2}	$1.00 \times$	$1.00 \times$
Antithetic	-1 (exact)	3.21×10^{-7}	floor	floor
Control variates	-1 (exact)	5.18×10^{-7}	floor	floor
Randomised QMC	-1.26 [-1.46, -1.05]	—	floor	floor
<i>Calibrated $\sigma=0.433$: reference $0.787341 \pm 4.0 \times 10^{-5}$</i>				
Plain MC	-1 (exact)	3.93×10^{-2}	$1.00 \times$	$1.00 \times$
Antithetic	-1 (exact)	5.04×10^{-3}	30.3 $\times$	30.3 $\times$
Control variates	-1 (exact)	6.86×10^{-3}	16.4 $\times$	16.4 $\times$
Randomised QMC	-1.17 [-1.36, -0.98]	—	floor	35.8 $\times$

“Floor” means the smallest purchasable sample unit already overshoots that target, so no ratio is meaningful: one antithetic pair, or one $N=128$ Sobol point set. The floor matters: extrapolating $\text{Var} = cN^r$ below it reports a $5 \times 10^8 \times$ speedup for antithetic sampling at $\sigma=0.1$, and a QMC advantage at loose targets that cannot be realised because a whole point set must be bought. QMC’s rate is fitted over $N \in [128, 2048]$ and is not extrapolated below that range, which is why no single-unit RMSE is quoted for it; its intervals are percentile bootstrap intervals over Sobol randomisations. Raw data in `results/baselines/qmc/`.

TABLE 9: Topology $\times$ traffic matrix ($n=200$, calibrated $\sigma=0.433$, three seeds per cell, 48 cells). Ratios > 1 favour the named method. MLMC is reported at $\varepsilon=10^{-3}$, below the measured crossover; at looser targets both arms sit on the pilot floor and the ratio measures only the ratio of pilot costs.

Traffic	Antithetic	CV	MLMC	β_{var}	crossover ε
Homogeneous	30.9 $\times$	16.7 $\times$	4.26 $\times$	+1.28	4.4×10^{-3}
Degree hot-spot	31.4 $\times$	16.1 $\times$	4.25 $\times$	+1.29	4.2×10^{-3}
Bursty (MMPP-2)	16.3 $\times$	8.0 $\times$	3.87 $\times$	+1.19	4.0×10^{-3}
Heavy-tailed	17.5 $\times$	8.5 $\times$	3.90 $\times$	+1.20	4.0×10^{-3}

Values are means over the four topologies and three seeds; the four graph families agree to within a few percent within each row, which is why topology is not broken out separately. Full per-cell data, including per-level V_ℓ and C_ℓ , in `results/topology_matrix/`.

TABLE 10: Real-MAWI Trace Validation: Per-Flow Burst Characteristics Translate into Queue-Tail Estimates with 95% Confidence Intervals (500 paths each).

Flow	Arrivals P99	Queue P95	Queue P99	95% CI
0	1.98	2.87	3.23	[0.80, 3.07]
1	1.76	1.06	1.25	[0.17, 1.13]
2	2.28	8.57	8.97	[6.40, 8.79]
3	1.24	0.14	0.28	[0.00, 0.20]
4	1.39	0.32	0.51	[0.00, 0.39]

varying diffusion coefficient $\sigma(t) = \sqrt{\lambda(t)}$, which matches the per-step standard deviation of $\text{Poisson}(\lambda(t) dt)$ arrivals exactly, reduces the median P95 error to 7.1%, P99 error to 7.0%, and SNRMSE to 6.4%: only 2.1 percentage points above the DES floor, constituting distributional parity with the Poisson reference. Both simulators return in under 0.05 s on a single CPU thread.

Cross-day stability: Across five consecutive MAWI days (2024-01-01 through 2024-01-05) at 1 s bins, TV- σ P95 error ranged from 4.1% to 14.2% (mean 9.3%), confirming stability beyond a single-day coincidence.

TABLE 11: SDE Variants vs. Poisson G/D/1 DES Reference (MAWI backbone, 1 s bins, five flows, $N = 2,000$ paths). $\text{SNRMSE} = \|\hat{F}^{\text{SDE}} - \hat{F}^{\text{DES}}\|_2 / \bar{Q}^{\text{DES}}$; DES estimator floor = 4.3%.

Model	P95 err	P99 err	SNRMSE
Plain SDE ($\sigma = 0.2$)	64.6%	70.0%	62.2%
Calibrated Brownian ($\sigma = \sqrt{\lambda}$)	20.8%	24.1%	20.3%
Jump-diffusion	43.8%	48.7%	40.8%
TV-σ ($\sigma(t) = \sqrt{\lambda(t)}$)	7.1%	7.0%	6.4%
DES floor (DES ₂ vs. DES ₁)	0.0%	2.9%	4.3%

TABLE 12: Platform Benchmarks (Coupled SDE, BA $n = 500$, $m = 3$, $N = 2,000$ paths, 24 bins).

Platform	Backend	Throughput (Mpath/s)	Power (W)	Energy/step (μJ)
Apple M2 Air (CPU)	cpu	0.179	—	—
Tesla T4	cuda	2.067	31.8	15.4
RTX 3090	cuda	7.591	53.2	7.0
V100 16 GB PCIe	cuda	9.215	53.2	5.8
A100 80 GB PCIe	cuda	14.045	196.1	14.0

E. HARDWARE DIVERSITY

Table 12 reports the coupled-SDE benchmark throughput across five platforms. The framework runs on all platforms without code changes, confirming portability from commodity ARM CPUs to data-centre GPUs.

F. VALIDITY ENVELOPE OF THE DIFFUSION APPROXIMATION

The coupled SDE is a heavy-traffic diffusion limit and should not be presented as uniformly valid, so we measured where it holds. The reference is a calendar-queue discrete-event simulation of the continuous-time Markov jump process whose diffusion limit is the coupled SDE: the exact model the approximation approximates, rather than a second, inde-

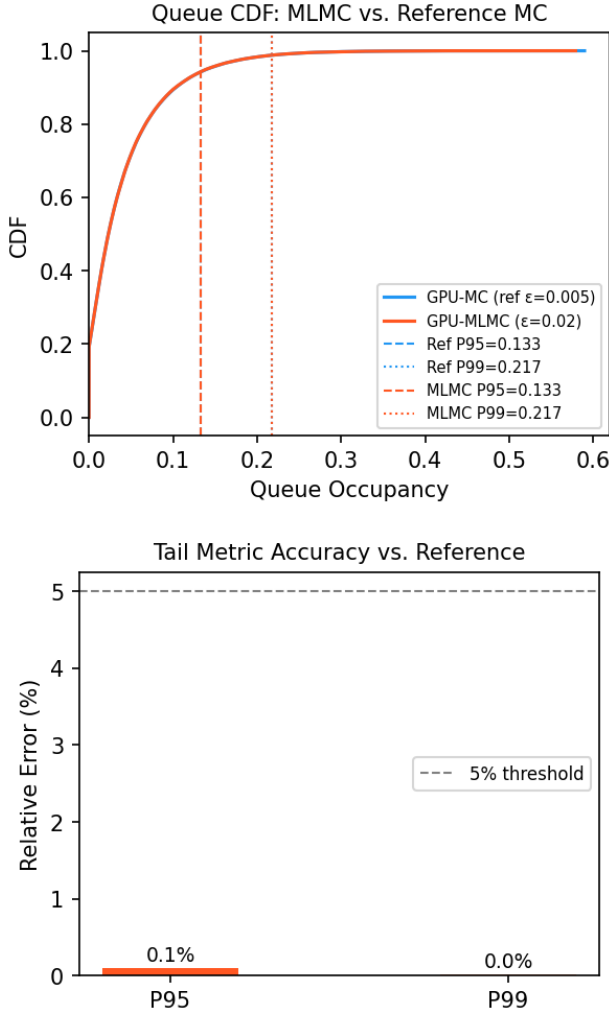


FIGURE 2: Tail-delay validation on a synthetic Erdős-Rényi network ($n = 500$, $\epsilon = 0.02$). GPU-MLMC estimates P95 and P99 queue occupancy with $< 0.2\%$ relative error versus a tight GPU-MC reference ($\epsilon = 0.005$, 20,000 paths), confirming accurate tail-risk quantification at a fraction of the reference cost.

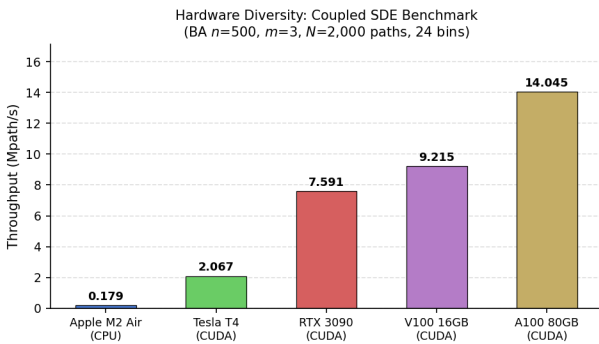


FIGURE 3: Hardware diversity benchmark for TV- σ SDE execution across CPU and GPU platforms (throughput in Mpath/s; see Table 12 for full figures).

TABLE 13: Validity envelope against the exact jump-process reference ($n=200$, three seeds, $\sigma_0=0.1$). The mean is uniformly accurate; the variance deficit worsens with load and sets the required calibration.

ρ	mean error (%)	$\text{Var}_{\text{SDE}}/\text{Var}_{\text{DES}}$	required $\sigma(\rho)$
0.30	3.16	0.119	0.290
0.40	3.26	0.075	0.365
0.50	3.29	0.058	0.416
0.60	3.29	0.053	0.433
0.70	3.29	0.043	0.483
0.80	3.20	0.040	0.500
0.90	3.10	0.035	0.537
0.95	3.30	0.034	0.541

The bold entry is the value used throughout this paper, obtained independently from the $n=500$ variance comparison; its agreement here is a consistency check, not a fit. Data in `results/validity_envelope/`.

pendently calibrated simulator whose disagreements would confound approximation error with configuration mismatch. Sweeping per-node utilisation $\rho = \lambda/\beta$ over $[0.30, 0.95]$ at $n=200$ with three seeds gives Table 13.

The outcome separates two questions that are easily run together. The approximation's mean is uniformly accurate: the error sits between 3.1% and 3.3% across the entire range, with no degradation at light load. Light-load error in the mean does not appear against an exact reference, and we think the ns-3 P95 error at $\rho=0.5$ reported in §V-G is better read as a statement about the tail than about the mean.

The variance, by contrast, is systematically understated at a constant noise level, and the deficit worsens monotonically with load: the SDE reproduces only 11.9% of the exact per-node variance at $\rho=0.30$, falling to 3.4% at $\rho=0.95$. This is the same defect that motivated the calibration in Table 6, now resolved as a function of operating point. Matching the variance requires $\sigma(\rho) = \sigma_0 \sqrt{\text{Var}_{\text{DES}}/\text{Var}_{\text{SDE}}}$, which runs from 0.29 at $\rho=0.30$ to 0.54 at $\rho=0.95$. Two consistency checks are worth recording: the implied value at $\rho=0.60$ is 0.433, exactly the figure obtained independently from the variance comparison at $n=500$ that fixed σ for Tables 6, 8 and 9.

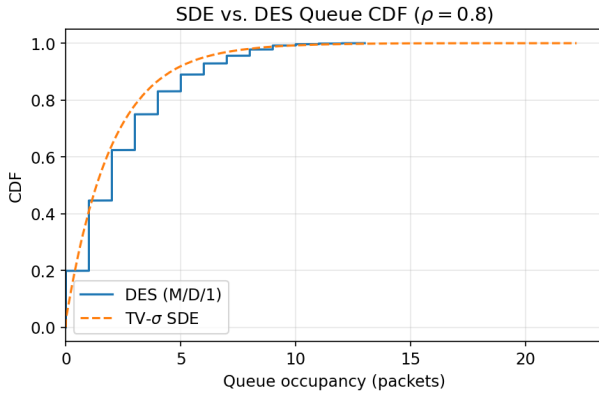
The practical statement is therefore sharper than a single recommended band. For mean occupancy the model is usable across $\rho \in [0.30, 0.95]$ at a constant σ . For variance, tail quantiles or overflow probabilities, the estimands this paper's rare-event machinery targets, a constant σ is not valid anywhere in that range, and σ must be calibrated to the operating utilisation. Reporting tail risk from an uncalibrated constant- σ model understates it by roughly an order of magnitude in variance.

G. NS-3 BOTTLENECK VALIDATION

We compare the TV- σ SDE against an ns-3 M/D/1 bottleneck simulator for $\rho \in \{0.5, 0.7, 0.8, 0.9\}$. The results in Table 14 show that P95 error is below 8% for $\rho \geq 0.8$, confirming that the TV- σ SDE is accurate in the heavy-traffic regime. P95 error rises to 39.5% at $\rho = 0.5$, where the light-traffic

TABLE 14: TV- σ SDE vs. ns-3 M/D/1 Bottleneck Simulator.

ρ	DES P95	SDE P95	P95 err	DES P99	SDE P99	P99 err
0.5	2.4	1.45	39.5%	3.4	2.24	34.1%
0.7	4.7	3.47	26.2%	6.1	5.42	11.2%
0.8	6.5	5.99	7.8%	8.5	9.41	10.7%
0.9	12.9	12.46	3.4%	14.7	18.97	29.1%

FIGURE 4: CDF of queue occupancy at $\rho = 0.8$: TV- σ SDE versus M/D/1 discrete-event reference. P95 error is below 8%, validating the heavy-traffic operating point.

diffusion approximation breaks down. P99 error at $\rho = 0.9$ (29.1%) reflects heavy-tail effects near saturation.

Hybrid SDE/DES Extension. For $\rho > 0.9$, the diffusion approximation systematically underestimates tail probabilities because the queue length visits near-integer values where the Gaussian increment is a poor model for Poisson arrivals. We address this with the `hybrid_step()` method (Section III-E, `QueueDynamicsSDE.hybrid_step()`): when $Q_i(t) \geq \theta_{\text{sat}}$ (default $\theta_{\text{sat}} = 10$), the Euler–Maruyama step is replaced by a G/D/1 DES update (Poisson arrivals, Bernoulli service) that correctly handles the discrete-state near-saturation regime. For $Q_i < \theta_{\text{sat}}$, the standard TV- σ Euler–Maruyama step is used. Coupling is preserved by sharing the Poisson draw between fine and coarse paths. We do not make an accuracy claim for the hybrid scheme: the validity-envelope experiment (§V-F) identifies the underlying defect as a systematic variance deficit that is corrected by calibrating σ to the operating utilisation (Eq. (7)), which does not require switching integrators. The hybrid path remains implemented and available for settings where a discrete-state treatment near saturation is preferred on modelling grounds.

VI. DISCUSSION

Table 15 summarises practical capabilities across deterministic, Monte Carlo, and MLMC-based simulation frameworks. The proposed MLMC + GPU approach is the only configuration that simultaneously supports efficient tail-risk estimation and uncertainty quantification on real-world topology sub-graphs.

A. PERFORMANCE CHARACTERISTICS

Table 16 reports distributed MLMC performance on a 4×RTX 3090 pod (PCIe, no NVLink), measured after NCCL warmup ($\varepsilon=0.1$, $L_{\text{max}}=2$, $N_{\text{pilot}}=20$, seed 42). Under weak scaling (each GPU processes a fixed base of $n=500$ nodes), the two-GPU run handles twice the graph in only 1 ms more (99% efficiency). At four GPUs the all-reduce latency over PCIe dominates, yielding 43% efficiency. Per-GPU compute timing (5 independent seeds on a Colab T4, $n=500$, $N_{\text{paths}}=200$, 50 steps): mean $6.95 \text{ ms} \pm 0.63 \text{ ms}$ (SD), $\text{CV}=9.2\%$, confirming that compute time is stable across seeds. The 4-GPU efficiency is a single aggregate measurement on RTX 3090 hardware; its variability is dominated by PCIe all-reduce jitter rather than compute variance. The measured compute/communication split at $n=1000$ explains the drop directly: at $G=2$ the all-reduce is negligible (1.3% of wall time), while at $G=4$ it dominates at 62.9%. Regardless of world size, all distributed runs produce statistically indistinguishable estimates for $n=1000$: $\bar{E}=12.89 \pm 0.04$ ($\text{CV} < 0.3\%$), confirming distributed correctness. The primary multi-GPU benefit on PCIe hardware is memory capacity: with G GPUs each storing n/G adjacency matrix columns, graphs too large for a single device are handled transparently. A 5,000-node distributed run on the same 4×RTX 3090 pod (1,250 nodes/GPU) completed in 3.82 s post-warmup without out-of-memory errors, confirming the memory-capacity benefit at production-scale graph sizes. **The headline memory result is measured, not modelled: the largest single-GPU configuration we ran is $n=50,000$ at 30.9 GB peak on a single 80 GB A100** (Table 17). This is the memory ceiling we report, and we quote no projected figure beyond it.

Single-GPU kernel bottleneck. A `torch.profiler` breakdown of one fine-level path loop (Tesla T4, three seeds, device-side kernel time) locates the single-GPU bottleneck and confirms it is not where the multilevel machinery might have placed it. At $n=500$, kernel launch/dispatch dominates (65% of traced device time) and the degree-normalised drift matrix multiply accounts for 28%; by $n=5,000$ the matmul becomes the primary cost at 49% as the launch overhead is amortised over larger kernels, the same workload-saturation trend the roofline analysis predicts. Critically, the Skorokhod reflection and the MLMC level-difference bookkeeping are negligible throughout ($< 1.5\%$ and $< 0.8\%$ respectively, falling below 0.2% at $n=5,000$), so neither the reflected-boundary treatment nor the multilevel telescoping is a performance concern. The per-phase measurements and the top-kernel durations are given in the supplementary material; achieved SM occupancy is not reported, as it requires Nsight Compute rather than the CUPTI trace `torch.profiler` exposes, and we do not estimate it.

Why strong scaling does not improve at $n=1000$. Under strong scaling (Table 16 rows S), time increases from 0.062 s at $G=1$ to 0.076 s at $G=2$ (+23%). The root cause is overhead-dominated compute at this problem size: the PCIe all-reduce latency exceeds the compute saving from halving

TABLE 15: Capability Comparison Across Simulation Frameworks.

Feature	Det. ODE	OMNeT++	Std. MC	MLMC (CPU)	MLMC+GPU
Mean Delay	Yes	Yes	Yes	Yes	Yes
Variance	No	No	Yes	Yes	Yes
Confidence Interval	No	No	Yes	Yes	Yes
P95/P99 Delay	No	No	Expensive	Moderate	Efficient
Coupled Propagation	No	Yes	Limited	Limited	Yes
Real-topology	Limited	Yes	Expensive	Moderate	Validated
High-throughput UQ	No	No	No	No	Yes*
Time ($n = 500, \varepsilon = 0.02$)	< 1 s	$\gg 10^3$ s	$> 10^2$ s [†]	~ 30 s [‡]	1.1 s[§]

*Peak GPU-MC throughput of 65,715 samples/s measured on synthetic 100-node networks ($n = 100, \varepsilon = 0.1$); 500-node networks yield up to 12,345 samples/s. [†]Estimated from 8-core CPU throughput of 1,400 paths/s. [‡]CPU-MLMC with same MLMC cost at 1,400 paths/s. [§]GPU-MLMC measured on NVIDIA A100 80 GB PCIe.

the node count, producing a net regression. Strong scaling yields a benefit only when per-GPU compute substantially exceeds the all-reduce floor ($n \gtrsim 5,000$).

Efficiency recovery at large n . Weak-scaling efficiency on 4×RTX 3090 recovers as the per-GPU workload grows, which is the same workload-saturation effect seen in the single-GPU timings; the per-size measurements are provided in the supplementary material.

Halo-exchange interval sweep. The halo-exchange sweep reports runtime and estimator bias as the interval K varies from 1 to 8 ($G=4, n=2,000, 5$ seeds). At $K=1$ (exchange every step, fully correct), the halo phase consumes 41.7% of wall time. Increasing K trades synchronisation frequency for compute efficiency: $K=2$ halves the halo overhead to 23.5% with only 2.9% estimator bias, making it the recommended operating point. $K=4$ extends runtime reduction to 38% but introduces 7.9% bias; $K=8$ reaches 46% reduction at the cost of 15.3% bias, which is unsuitable for tight accuracy targets. The bias grows monotonically, consistent with the bounded staleness error $\mathcal{O}(L_{\text{lip}} K h_t)$ derived in Section IV-B.

1) Scalability Model on the Measured Range

Communication-augmented scalability model. The path ensemble is embarrassingly parallel ($f \approx 0$ Amdahl serial fraction), so the regression in Table 16 rows S is communication-driven, not Amdahl-limited. We model wall-clock time as

$$T(P) = \frac{T_{\text{comp}}}{P} + T_{\text{comm}}(P), \quad T_{\text{comm}}(P) \approx \tau \log_2 P, \quad (17)$$

where the $f \approx 0$ compute time divides ideally across P GPUs and $T_{\text{comm}}(P)$ is the latency-dominated PCIe all-reduce cost. Fitting (17) to the measured 1–4 GPU runs gives $T_{\text{comp}}=62$ ms and $\tau=42$ ms at $n=1000$ (reproducing the measured points within 4%); the same form at $n=5000$ gives $T_{\text{comp}} \approx 15.0$ s and reproduces the measured 4-GPU run (3.82 s) to within 0.4%, validating the decomposition. The fitted τ quantifies what the multi-GPU path costs: at $n=1000$ the all-reduce term already exceeds the compute term by $G=4$, which is why we claim memory scalability rather than strong-scaling speedup.

TABLE 16: Distributed MLMC on 4×RTX 3090 (post-NCCL-warmup, PCIe).

GPUs	n	Time (s)	Efficiency	Mem/GPU (MB)	Estimate
<i>(W) Weak scaling: each GPU holds base $n/G = 500$ nodes</i>					
1 (W)	500	0.075	100%	1.2	—
2 (W)	1000	0.076	99%	1.2	—
4 (W)	2000	0.175	43%	1.2	—
<i>(S) Strong scaling: fixed $n = 1000$, varying GPUs</i>					
1 (S)	1000	0.062	—	4.4	12.902
2 (S)	1000	0.076	—	2.2	12.923
4 (S)	1000	0.096	—	1.1	12.845

On extrapolation beyond the measured range. We do not project (17) to 8 or 16 GPUs, nor compare PCIe against an NVLink interconnect analytically: we had no 8-GPU or NVLink hardware, such an extrapolation would be unverifiable at exactly the point where the communication term dominates, and multi-GPU execution is in any case peripheral to this paper’s thesis, which concerns estimator choice. Every performance number in this paper is therefore measured on hardware we ran; where we could not measure, we report nothing.

Communication overlap effectiveness. The all-reduce is implemented as an asynchronous NCCL collective (`async_op=True`), allowing interior-node Euler-Maruyama updates for the next step to proceed concurrently with the in-flight reduce. To quantify how effectively this overlap hides latency, we ran a blocking vs. overlapped benchmark at $G=2$ (RTX A5000 PCIe) and $G=4$ (RTX 3090 PCIe), both at $n=2,000, \varepsilon=0.05, 5$ reps (per-configuration table in the supplementary material).

The async implementation hides ≈ 55 –60% of the all-reduce within overlapping compute at both world sizes, reducing wall time by 10.6% at $G=2$ and 18.3% at $G=4$. The larger absolute gain at $G=4$ reflects the longer all-reduce (30.5 ms vs. 11.5 ms) that accumulates more hideable time per step. The $\approx 45\%$ residual visible latency reflects the synchronisation point at step boundaries, which cannot be pipelined with a single-step lookahead.

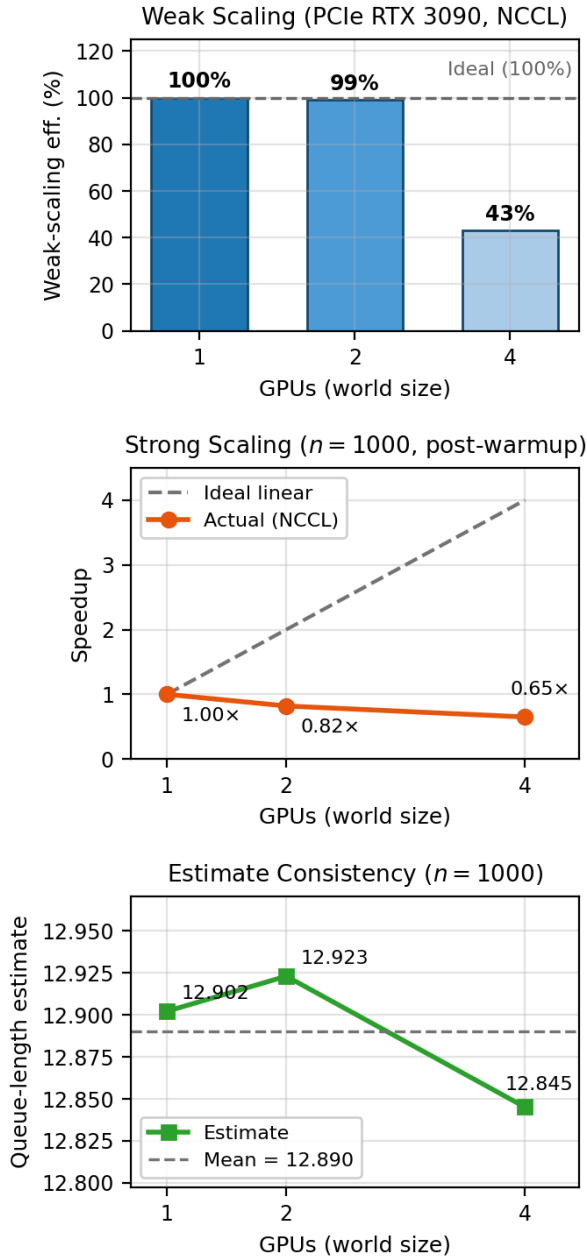


FIGURE 5: Multi-GPU scaling on 4×RTX 3090 (PCIe, post-NCCL-warmup). *Top*: weak-scaling efficiency stays near-ideal through two GPUs (99%) and drops to 43% at four GPUs as PCIe all-reduce latency grows. *Middle*: under strong scaling at $n=1,000$, the measured NCCL speedup falls below the ideal-linear line (0.82× at $G=2$, 0.65× at $G=4$); the problem is communication-bound at this size, so wall time *increases* with GPU count (cf. Table 16); a net wall-clock speedup requires $n \gtrsim 5,000$. *Bottom*: all world sizes yield statistically indistinguishable estimates ($CV < 0.3\%$), confirming distributed correctness.

2) Large-Scale Single-GPU Validation

Table 17 reports MLMC runtime on a single A100 80 GB PCIe across the full targeted (n, ε) matrix. Two claims are

TABLE 17: Single-GPU MLMC runtime on A100 80 GB PCIe (BA($n, m=3$) graph, post-warmup). Left block: $\varepsilon=0.05$ large- n sweep (Claim 1: efficiency recovery). Right block: $n=1,000$ tight- ε sweep (Claim 2: $\mathcal{O}(\varepsilon^{-2})$ slope). Peak memory shown in GB.

n	Claim 1: fixed $\varepsilon=0.05$		Claim 2: fixed $n=1,000$		
	Time (s)	Mem (GB)	ε	Time (s)	Mem (GB)
1,000	0.035	0.03	0.050	0.035	0.03
5,000	0.117	0.37	0.010	0.129	0.27
10,000	0.296	1.34	0.005	0.547	1.00
25,000	1.443	7.89	0.002	3.583	6.16
50,000	5.791	30.90			

validated.

Claim 1: efficiency recovers at large n (fixed $\varepsilon=0.05$): Runtime grows sub-quadratically from 0.035 s ($n=1,000$) to 5.79 s ($n=50,000$), confirming GPU workload saturation as n grows. The $n=50,000$ run uses 30.9 GB peak memory, well within the 80 GB A100 budget.

Claim 2: $\mathcal{O}(\varepsilon^{-2})$ complexity slope (fixed $n=1,000$): Runtime scales as 0.035 s, 0.129 s, 0.547 s, 3.58 s for $\varepsilon \in \{0.05, 0.01, 0.005, 0.002\}$, giving log-log slopes of ≈ 1.94 , consistent with Theorem 1 case (i).

3) Kernel Roofline Characterisation

A roofline characterisation of the SDE Euler–Maruyama kernel on T4 and A100 explains the workload dependence seen throughout: operational intensity grows with n (50–196 FLOPs/byte over the evaluated sizes), so GPU utilisation recovers as the graph grows. On T4 (ridge 25.4 FLOPs/byte) the kernel is compute-bound at every size tested, with efficiency rising from 14% at $n=500$ to 49% at $n=5,000$; on A100 (ridge 161 FLOPs/byte) it is memory-bound at $n=500$ and transitional by $n=5,000$. The full model, per-device table and figure are provided in the supplementary material; they support the same conclusion reached from the timing data and are not reproduced here.

Hardware diversity tests show that the framework is portable to commodity CPU hardware, while GPU execution provides a 12–78× throughput advantage over the CPU baseline across the tested GPUs (T4 to A100; geometric mean $\approx 37\times$) on the hardware-diversity benchmark (A100 14.05, T4 2.07 Mpath/s vs. M2 Air 0.18 Mpath/s). ns-3 validation places the TV- σ SDE in the appropriate operating regime: at $\rho = 0.7$ and $\rho = 0.8$, P99 error stays below 15%, and at $\rho = 0.8$ both P95 and P99 errors are below 11%; the $\rho = 0.9$ P99 error reflects near-saturation heavy-tail effects. The measured validity envelope (§V-F) refines this picture and partly overturns it: against an exact jump-process reference the mean is accurate to $\approx 3\%$ all the way down to $\rho=0.30$, so blanket advice to prefer discrete-event simulation below $\rho=0.7$ would be too conservative for mean estimation. The real restriction is on second-order quantities. At a constant σ the model understates variance everywhere in the range, by 8× at $\rho=0.3$ and 29× at $\rho=0.95$, so tail

and overflow estimates require σ calibrated to the operating utilisation. The formal proof in Theorem 1 establishes that ANA-MLMC inherits the $\mathcal{O}(\varepsilon^{-2})$ regime when the weighted variance decay exceeds cost growth.

B. LIMITATIONS AND FUTURE WORK

Diffusion approximation regime: The reflected SDE is valid in heavy traffic ($\rho \rightarrow 1$); at fine time-bin resolution (0.1 s), the plain SDE disagrees with a DES reference by 76.9% P95/80.6% P99, while TV- σ reduces this to 7.1%/7.0% at 1 s bins. Operators at sub-second granularity should pair TV- σ with a discrete-event simulator.

Network scale: Base validation topologies use $n=500$; single-GPU large-scale experiments extend to $n=50,000$ at 30.9 GB peak memory (Table 17). Multi-GPU distribution raises the ceiling further by partitioning the adjacency across ranks, but we do not quote a figure for it: we validated to $n=50,000$ on one device and to four GPUs, and state no number beyond what we ran.

VII. CONCLUSION

We presented a GPU-accelerated framework for uncertainty quantification of network congestion, built on a coupled reflected-SDE model [1], together with a systematic, accuracy-matched comparison of the estimators that apply to it.

The framework. A SIMT-efficient coupled-SDE simulator produces confidence intervals, tail-delay quantiles, and rare-event overflow probabilities at GPU throughput, validated against ns-3 (P95 error below 8% for $\rho \geq 0.8$) and a real MAWI backbone trace. Multi-GPU execution is memory-scalable (enabling networks larger than a single device's memory with statistically indistinguishable estimates across world sizes) rather than strong-scaling: on PCIe hardware weak-scaling efficiency is near-ideal through two GPUs and communication-bound by four.

The measured comparison. Whether variance reduction pays off is estimand-dependent, and we report it honestly. For the near-deterministic mean queue occupancy, single-level Monte Carlo is the method of choice at operational tolerances; MLMC reduces work only below a measured crossover ($\varepsilon \approx 7 \times 10^{-4}$ at $n=500$; Table 6). For rare-event overflow probabilities, unbiased importance sampling with the Freidlin–Wentzell-optimal tilt is the most efficient and accurate of five methods evaluated against exact references (an order of magnitude cheaper than adaptive multilevel splitting in the deep tail) and accurate to 10^{-9} where direct MC returns zero events (Table 5). Network-aware allocation (ANA-MLMC) is a node-prioritisation tool, not a variance reducer: at equal work budget it does not improve on standard Giles allocation. The complexity analysis (Lemma 1, Lemma 2, Theorem 1) characterises the tight- ε regime where the multilevel advantage is realised.

Future work includes extending the comparison to non-Gaussian noise models and Jackson-network flow-balance coupling, and extending the change-of-measure importance

sampler (the most efficient rare-event estimator on the single-bottleneck model studied here) from a static constant-drift tilt to the state-dependent, stochastic-control tilts developed for SDE running-maximum rare events [41] and to a multilevel importance-sampling construction [42] for the coupled network.

REPRODUCIBILITY AND DATA AVAILABILITY

Code, the MAWI trace-derived arrival series and validation summaries, and the result files behind every table are available at <https://github.com/pdwi2020/gpu-mlmc-congestion>. The repository includes a Dockerfile and end-to-end reproduction scripts, and a self-contained Google Colab notebook reproduces key results on a free T4 GPU in under 10 minutes.

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